

**National Oceanic and Atmospheric Administration (NOAA)
National Marine Fisheries Service (NMFS)
Southwest Fisheries Science Center (SWFSC)
Marine Mammal and Turtle Division (MMTD)**

**STATEMENT OF WORK
Genetic Stock Identification of Loggerhead and Green Turtles**

I. Background/Overview:

NOAA's management of sea turtles, particularly under the Endangered Species Act (ESA), and fisheries under the Magnuson-Stevens Act (MSA) fundamentally depends on accurately identifying the genetic stock origin of turtles in US waters at the finest scale population Assessment Unit possible [e.g., Distinct Population Segment (DPS), Management Unit (MU), Recovery Unit (RU)]. Achieving this high-resolution Genetic Stock Identification (GSI), requires the use of mitochondrial DNA (mtDNA) markers combined with informative nuclear DNA (nDNA) markers (e.g. microsatellites and SNPs), for a mixed-marker approach—to better delineate Assessment Units (like DPS, MU, and RU) and facilitate the effective recovery and management of listed sea turtle species.

The SWFSC conducts GSI of sea turtles collaboratively with NMFS Science Centers and continues to develop and apply new genomic markers tailored to species and regions, with notable progress with Pacific and Atlantic leatherback turtles. However, there is a notable gap in available genetic data from key populations of Atlantic green and loggerhead turtles, and there is an immediate need to generate data for a backlog of samples at SWFSC using informative mtDNA and nDNA markers to allow GSI in order to achieve current science and management objectives identified under the NMFS protected species Stock Assessment Improvement Plan (SAIP). The newly proposed Critical Habitat for green turtles in waters surrounding these US territories (88 FR 46572) elevates the need to improve the genetic nesting population baseline dataset to improve capacity to determine connectivity with critical habitat at the DPS scale.

II. Scope of Work

This project will generate mtDNA and nDNA data for samples obtained by SWFSC to fill in critical knowledge gaps for Atlantic green and loggerhead turtles, using customized arrays of markers that are compatible with published and in-progress studies. These include standard mtDNA Control Region (CR) sequences for all samples, plus additional informative mtDNA markers for green turtles, and arrays of nDNA microsatellite markers individually customized for each species. The data produced will allow GSI of loggerhead bycatch and relocation trawl sample sets, fine-scale stock structure analysis of Atlantic green turtles, connectivity of green turtles foraging in critical US habitat, and analysis to delineate Atlantic DPS boundaries, all SAIP priorities that will inform current and upcoming NMFS Biological Opinions and population assessments.

III. Required Tasks

- **Task 1:** DNA extraction and Quality Control/assessment.
- **Task 2:** Sequencing of standard mtDNA CR fragment (~850 bp).
- **Task 3:** Sequencing of green turtle informative mitogenomic (mtDNA SNP) markers.
- **Task 4:** Genotyping of loggerhead turtle samples with custom microsatellite (nDNA) array of 16 loci.
- **Task 5:** Genotyping of green turtle samples with custom microsatellite (nDNA) array of 13 loci.
- **Task 6:** Deliver sequence data as FASTQ files and genotype data as allele calls to the Technical Point of Contact (TPOC).
- **Task 7:** Process up to 740 samples including but not limited to, loggerhead turtle genotyping, green turtle genotyping, green turtle control region sequences, and green turtle mitogenomic SNPs.

IV. Required Knowledge and Expertise

- Expertise in sanger sequencing techniques and microsatellite multi-locus genotyping.
- Expertise in Quality Assessment/Quality Control and processing of sequencing data.
- Expertise using Genemapper workflow system.
- Knowledge of informative mitogenomic markers appropriate for Atlantic green turtle GSI
- Knowledge of informative microsatellite markers appropriate for Atlantic green turtle and Atlantic loggerhead GSI
- Demonstrated knowledge of optimized multi-locus microsatellite panel and operational protocols for informative Atlantic loggerheads and green turtle GSI
- Access to laboratory and computing facilities for the above analyses, and ability to provide access to the raw data.

V. Period of Performance

The period of performance for this requirement is one (1) base year and four (4) option years. Current dates are estimates only. Period of Performance dates will be solidified closer to contract award.

Base Year: 2026 - 2027

Option Year 1: 2027 - 2028

Option Year 2: 2028 - 2029

Option Year 3: 2029 - 2030

Option Year 4: 2030 - 2031

VI. Place of Performance

The performance of these contracted activities will take place off-site at the contractors location.

VII. Deliverables Schedule

Item	Description	Due Date	Deliver To	Reference
Raw Sequences (FASTQ Files)	Sequence data delivered as FASTQ files	Upon completion	TPOC	SOW Section III, Task 6
Genotype Data	Genotype data delivered as allele calls	Upon completion	TPOC	SOW Section III, Task 6
NOAA Sexual Assault and Sexual Harassment Training (SASH)	The contractor shall provide all contractor employees assigned to perform under this contract with mandatory sexual assault and sexual harassment prevention and response training	Within 30 Days After Award	CO	NAM 1330-52.222-70